

External Interrupt Experiment

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1. Learning Objectives

1. Learn about external interrupt knowledge on the MSPM0G3519 motherboard.
2. Use the onboard button to control the onboard LED light on/off.

2. External Interrupt

An interrupt is when the CPU, while executing a program, has another event that needs to occupy the CPU. After receiving this interrupt request, the CPU pauses the execution of the original program to execute the interrupt function. After the interrupt function completes, it continues to execute from where it left off in the original program.

External Interrupt refers to an interrupt triggered by external hardware events (usually signals from outside the microcontroller). This interrupt can interrupt the current program execution flow of the microcontroller, allowing it to respond quickly to external hardware events.

External interrupts are usually associated with specific pins (GPIO pins), such as buttons, sensor signals, timer signals, etc. These pins are typically configured to detect specific signal changes (such as rising edge, falling edge, or level changes), thereby triggering an interrupt.

External Interrupt Advantages and Functions

Advantages:

- **Fast Response:** External interrupts can respond immediately when a triggering event occurs, without polling signals.
- **Resource Saving:** No need to continuously detect signal status in the main loop, saving CPU resources.
- **Improved Real-time Performance:** The interrupt mechanism ensures timely processing of critical events, thereby improving system real-time performance.

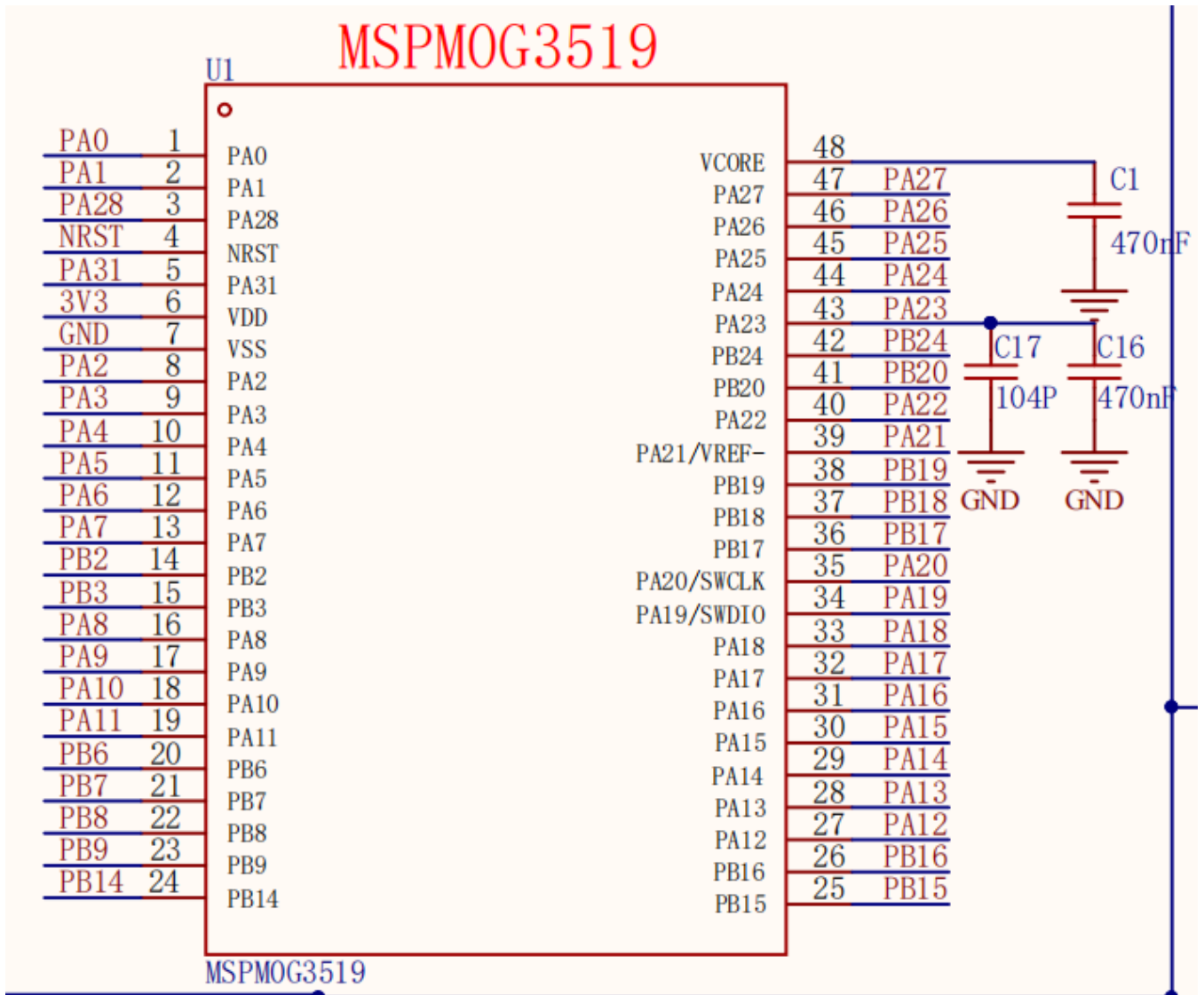
Functions:

- **Fast Response to External Events:** Can immediately interrupt the current program execution and quickly process the event when hardware events occur, without polling signal status. Avoids the program polling for events, greatly saving CPU resources and improving the system's response speed to external events.
- **Improved System Real-time Performance:** In scenarios requiring real-time response, external interrupts ensure critical events are processed in a timely manner without waiting for the system main program to complete its current task, thereby improving system real-time performance.
- **Reduced Main Program Burden:** External interrupts separate tasks related to external events from the main program, avoiding polling signal status in the main program and improving main program execution efficiency, allowing the main program to focus on core tasks.
- **Core of Event-Driven Systems:** Can implement efficient event-driven mechanisms and reduce system complexity.
- **Hardware Interaction and Control:** External interrupts enable microcontrollers to real-time process and respond to external hardware signals, ensuring the system operates normally.
- **Data Acquisition and Processing:** In many real-time data acquisition systems, external interrupts are used to trigger data acquisition or processing.
- **Low-Power Scenario Triggering Mechanism:** Through external interrupts, efficient low-power design can be achieved, allowing the system to remain in sleep state when not working and only run when necessary.
- **System Security Assurance:** Timely processing of security-related events to ensure system reliability and safety.

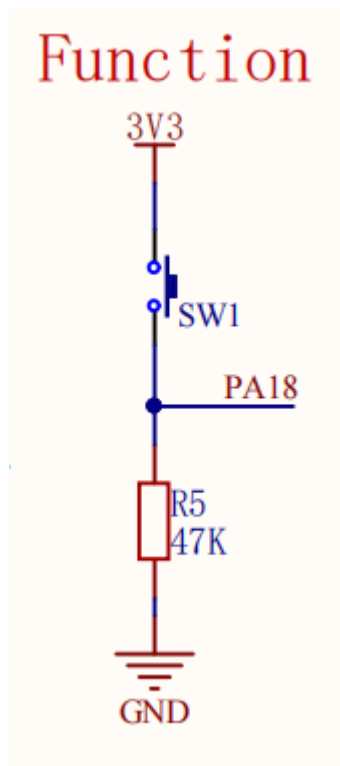
3. Hardware Setup

This course does not require additional hardware. You can directly use the onboard function button on the MSPM0G3519 development board.

MSPM0G3519 Main Control Diagram



Button Part Schematic

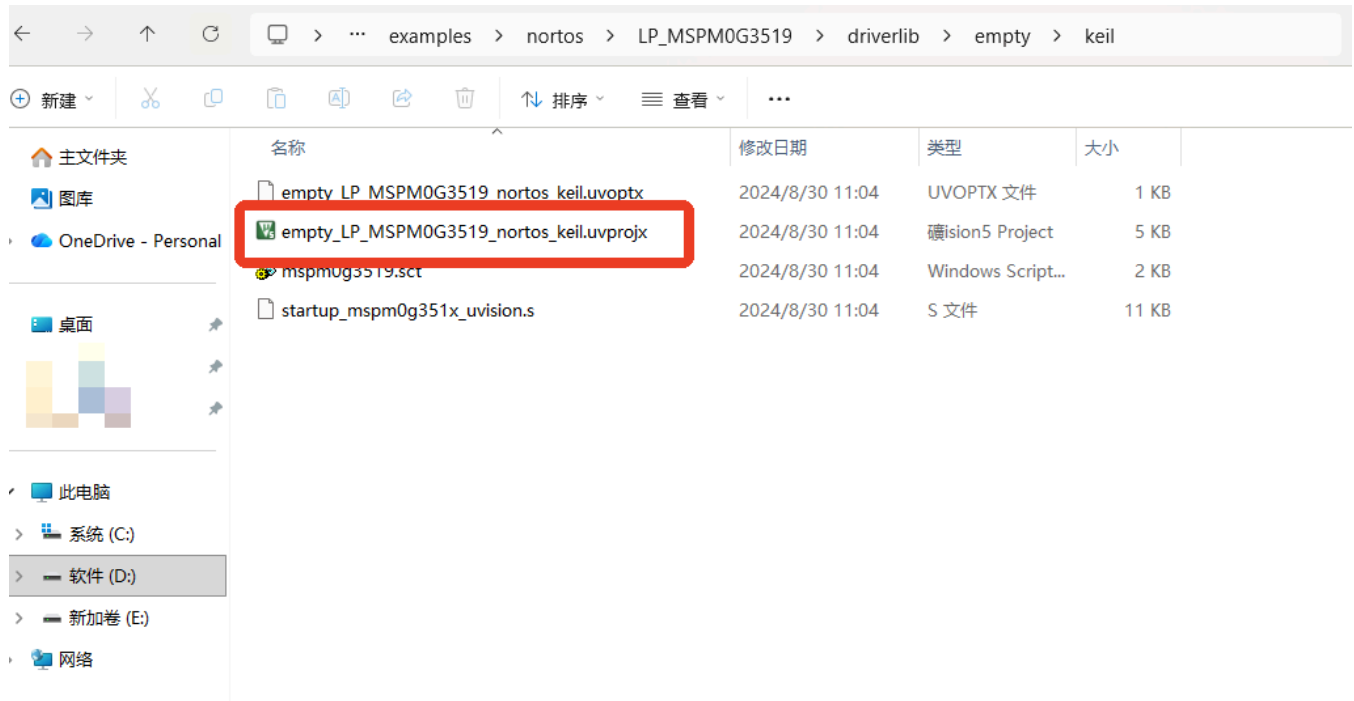


4. Experiment Steps

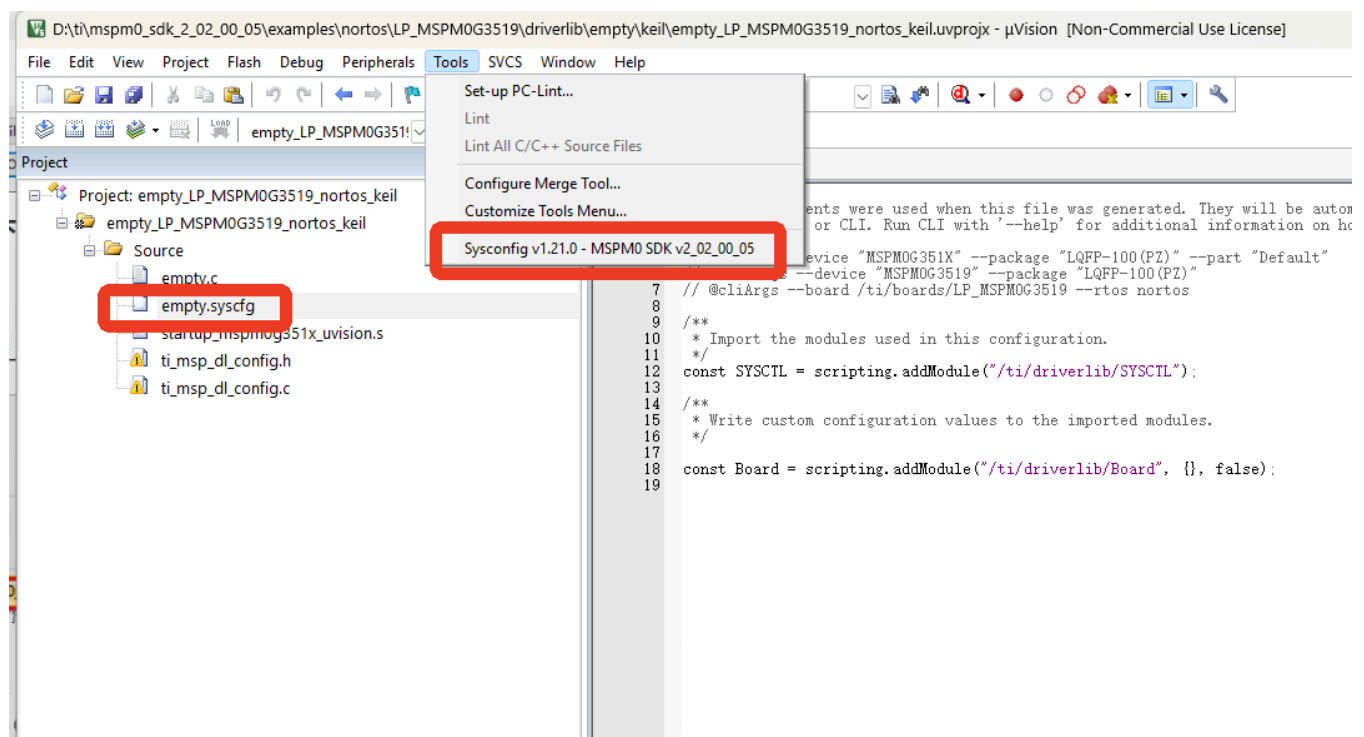
In this course, the PA18 button will be used to trigger LED1 on PB2 to light up and go out. In software, CPU interrupt mode is used. For LED1 PB2 pin configuration, please refer to the **Turn On LED Light** course.

1. Open SYSCONFIG Configuration Tool

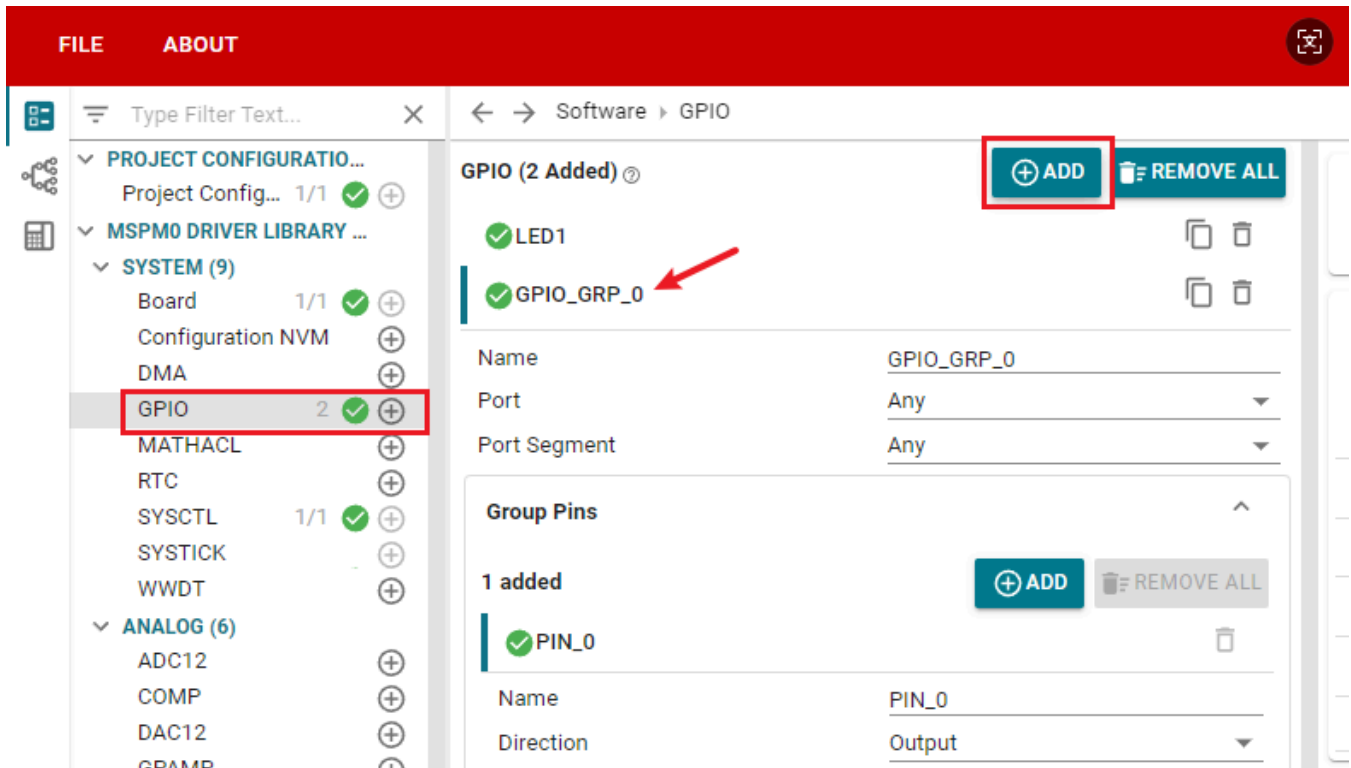
Open the empty project from the SDK in KEIL.



After selecting and opening it, open the empty.syscfg file in the KEIL interface. **With the empty.syscfg file open**, then open the SYSCONFIG GUI interface from the Tools menu.



In SYSCONFIG, select MCU peripherals on the left, find the GPIO option and click to enter. Click ADD in GPIO to add a GPIO group. This GPIO_GRP_0 is the newly added one.



2. Configure Pin Mode

Configure pin parameters, name the GPIO instance `KEY`. Since it is connected to PA18, Port selects `PORTA`.

Set the pin name to `PIN_18`, set to input mode `Input`. Internal Resistor sets to pull-down resistor `Pu11-Down Resistor`. Assigned Pin sets the controlled pin to 18.

✓ KEY

Name	KEY
Port	PORTA
Port Segment	Any

Group Pins

1 added

+

 ADD

REMOVE ALL

✓ PIN_18

Name	PIN_18
Direction	Input
IO Structure [?]	Any

Digital IOMUX Features

Internal Resistor	Pull-Down Resistor
Invert	Disabled
Input Filter	Disabled
Enable Fast-Wake	☐
Override Pin Hysteresis Default	☐
Hysteresis Control	Disabled
Wakeup Logic	Disabled

Assigned Port	PORTA
Assigned Port Segment	Any
Assigned Pin	18

Interrupts/Events

LaunchPad-Specific Pin	No Shortcut Used
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PinMux Peripheral and Pin Configuration

PIN_18	Any(PA18/11)
--------	--------------

Configure pin external interrupt. All GPIOA ports brought out from the chip can have external interrupt functionality set.

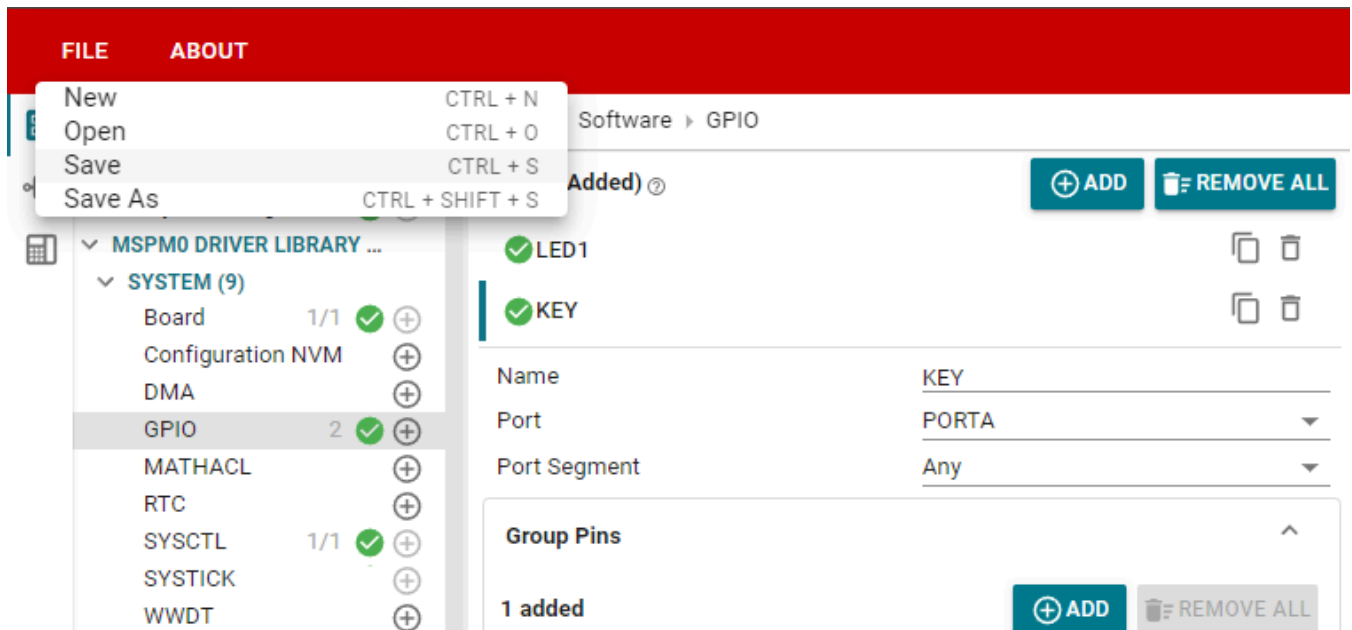
Enable Interrupts enables the interrupt. **Interrupt Priority** sets the interrupt priority to default. When multiple interrupts trigger simultaneously, the chip's default priority rules are followed.

Trigger Polarity sets the interrupt trigger mode to rising edge trigger. When the button is pressed, the GPIO pin level changes from low (0) to high (1). This process is called a rising edge. This means the external interrupt will be triggered when the pin state changes from low to high.

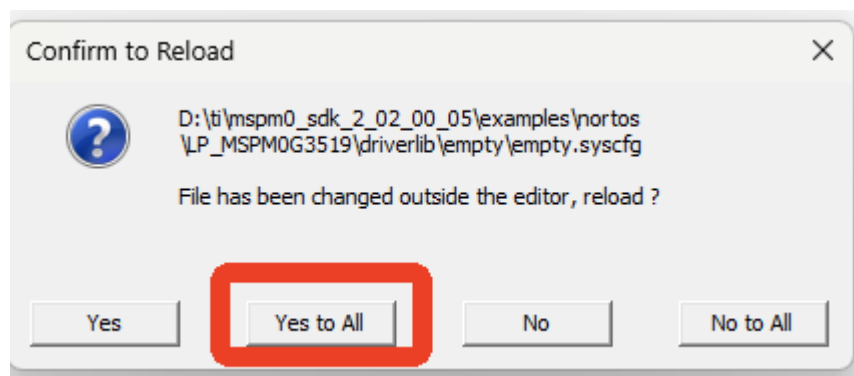
Event Publishing Channel does not enable the event publishing channel. The event publishing channel is a mechanism for publishing interrupt events to specified channels. Other peripherals can respond by subscribing to events on the channel.

Interrupts/Events	
Enable Interrupts	☒
Interrupt Priority	Default
Trigger Polarity	Trigger on Rising Edge
Event Publishing Channel	Disabled (0)

Click SAVE to save the SYSCONFIG configuration, then close SYSCONFIG and return to KEIL.



Select **Yes to All** in the popup.



We also need to confirm whether `ti_msp_dl_config.c` and `ti_msp_dl_config.h` files are updated. Simply compile, and the compilation will automatically update to KEIL. If not updated, we can also copy the file content from SYSCONFIG.

3. Write Program

After configuring the pin interrupt, we also need to manually write the external interrupt service function. Because we enabled the pin falling edge interrupt, when our button goes from pressed to released, the generated falling edge will trigger an interrupt, and after triggering, it will execute the interrupt service function. The interrupt service function name for each interrupt is usually fixed and cannot be modified arbitrarily; otherwise, you will not be able to correctly enter the interrupt service function. The specific name of the interrupt service function can be found in the project's `startup_mspm0g350x_uvision.s` file, which lists the interrupt service function names for each interrupt source.

```

startup_mspm0g350x_uvision.s
67
68         AREA      RESET, DATA, READONLY
69         EXPORT    __Vectors
70         EXPORT    __Vectors_End
71         EXPORT    __Vectors_Size
72
73         __Vectors  DCD      __initial_sp          ; Top of Stack
74                   DCD      Reset_Handler         ; Reset Handler
75                   DCD      NMI_Handler           ; NMI Handler
76                   DCD      HardFault_Handler     ; Hard Fault Handler
77                   DCD      0                     ; Reserved
78                   DCD      0                     ; Reserved
79                   DCD      0                     ; Reserved
80                   DCD      0                     ; Reserved
81                   DCD      0                     ; Reserved
82                   DCD      0                     ; Reserved
83                   DCD      0                     ; Reserved
84                   DCD      SVC_Handler           ; SVCcall Handler
85                   DCD      0                     ; Reserved
86                   DCD      0                     ; Reserved
87                   DCD      PendSV_Handler        ; PendSV Handler
88                   DCD      SysTick_Handler       ; SysTick Handler
89
90         ; External Interrupts
91         DCD      GROUP0_IRQHandler              ; 1:  GROUP0 interrupt handler
92         DCD      GROUP1_IRQHandler              ; 2:  GROUP1 interrupt handler
93         DCD      TIMG8_IRQHandler               ; 3:  TIMG8 interrupt handler
94         DCD      UART3_IRQHandler               ; 4:  UART3 interrupt handler
95         DCD      ADC0_IRQHandler                ; 5:  ADC0 interrupt handler

```

According to the MSPM0G series user manual, after GPIO interrupts are triggered, they are all published to the bus through the **GRP1** line. After the bus recognizes it, it enters the interrupt service function to execute the content.

So in the empty.c file, write the following code

```

#include "ti_msp_dl_config.h"

int main(void)
{
    SYSCFG_DL_init();
    NVIC_EnableIRQ(KEY_INT_IRQN); //Enable GPIOA port interrupt for button pin

    while (1)
    {

    }
}

void GROUP1_IRQHandler(void) //Interrupt service function of Group1
{
    // Read the interrupt register of Group1 and clear the interrupt flag
    switch( DL_Interrupt_getPendingGroup(DL_INTERRUPT_GROUP_1) )
    {
        // Check if it is the GPIOA port interrupt of KEY, note that it is INT_IIDX, not
        PIN_18_IIDX
        case KEY_INT_IIDX:
            // If the button is pressed, it becomes high level
            if( DL_GPIO_readPins(KEY_PORT, KEY_PIN_18_PIN) > 0 )
            {

```

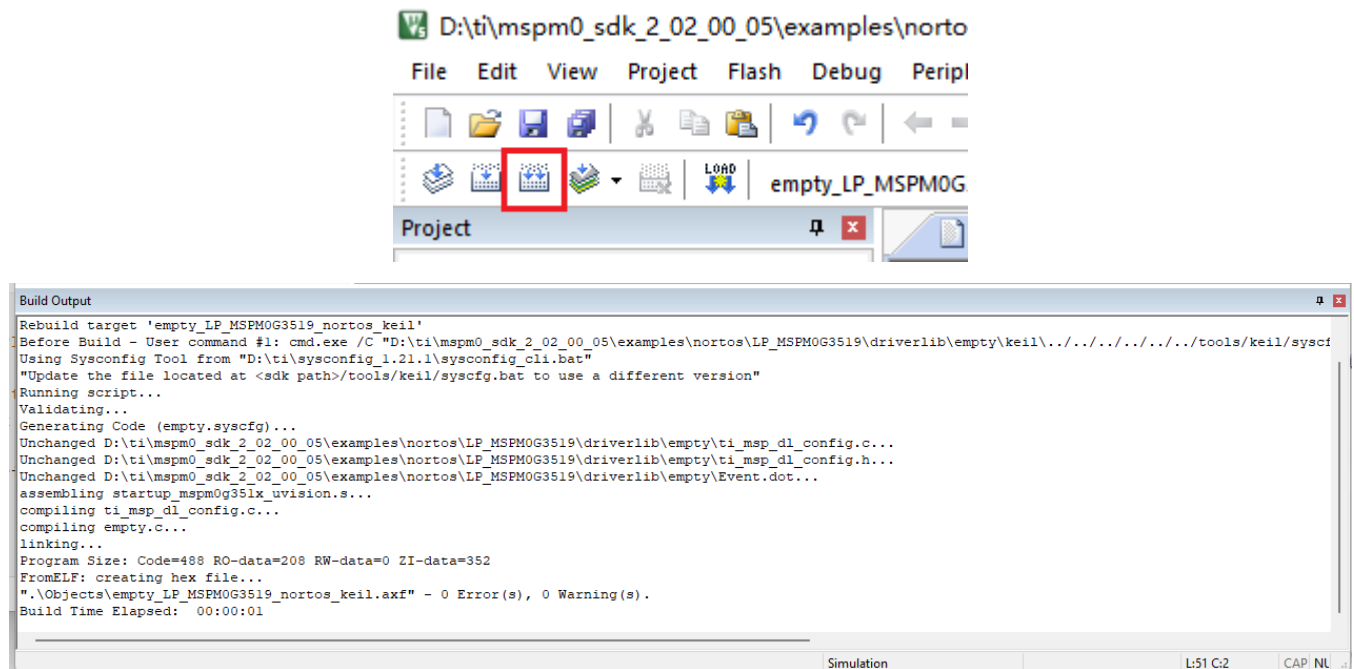
```

        // Set LED pin state toggle
        DL_GPIO_togglePins(LED1_PORT, LED1_PIN_2_PIN);
    }
    break;
}
}
}

```

4. Compile

Click the Rebuild icon. The following prompt indicates compilation is complete with no errors.



4. Program Analysis

- dl_gpio.h

```

/**
 * @brief      Get highest priority interrupt pending in the selected interrupt
 *             group
 *
 * @param[in]  group  Group of interrupts to check
 *
 * @return     The highest priority pending interrupt for the group requested.
 *             One of @ref DL_INTERRUPT_GROUP_IIDX
 */
__STATIC_INLINE uint32_t DL_Interrupt_getPendingGroup(DL_INTERRUPT_GROUP group)
{
    return (CPUSS->INT_GROUP[group].IIDX);
}

```

uint32_t DL_Interrupt_getPendingGroup(DL_INTERRUPT_GROUP group) is used to get the highest priority pending interrupt index in the specified interrupt group.

group: An enumeration type `DL_INTERRUPT_GROUP` used to specify which interrupt group to check.

- empty.c

```
int main(void)
{
    SYSCFG_DL_init();
    NVIC_EnableIRQ(KEY_INT_IRQN); //Enable GPIOA port interrupt for button pin

    while (1)
    {

    }
}

void GROUP1_IRQHandler(void) //Interrupt service function of Group1
{
    // Read the interrupt register of Group1 and clear the interrupt flag
    switch( DL_Interrupt_getPendingGroup(DL_INTERRUPT_GROUP_1) )
    {
        // Check if it is the GPIOA port interrupt of KEY, note that it is INT_IIDX, not
        PIN_18_IIDX
        case KEY_INT_IIDX:
            // If the button is pressed, it becomes high level
            if( DL_GPIO_readPins(KEY_PORT, KEY_PIN_18_PIN) > 0 )
            {
                // Set LED pin state toggle
                DL_GPIO_togglePins(LED1_PORT, LED1_PIN_2_PIN);
            }
            break;
    }
}
```

This code implements a basic interrupt service function, mainly using button-triggered interrupts to control the status of an LED.

After the `main()` function initializes, it enables the GPIOA port interrupt for the button. The `GROUP1_IRQHandler()` interrupt service function gets the highest priority pending flag in Group1, checks if a button interrupt occurred, then calls `DL_GPIO_readPins(KEY_PORT, KEY_PIN_18_PIN)` to read the button pin level change. If it becomes high level, the LED pin state toggles.

5. Experimental Phenomena

After the program is downloaded, you can control the LED light status with the button. Press once, the LED turns on; press again, the LED turns off.